Drumstick controlled percussion synthesizer proposal

Project concept

Our team plans to create an interactive percussion instrument that combines physical striking with synthesized sounds. The prototype consists of a drumstick with a built-in motion sensor, a small wooden striking plate equipped with a piezoelectric contact pickup, and a Daisy Seed control unit. When the user strikes the plate, the system captures the vibrations and triggers an electronic sound; the force of the strike determines the intensity of the audio response, while tilting the drumstick alters the pitch.
We aim to design an instrument that encourages effortless exploration during performance, allowing users to intuitively perceive how changes in movement affect the sound. This project will investigate how striking and tilting actions can be used to achieve clear, repeatable control over synthesized percussion timbres.

Interaction and listening experience

User action	Planned sound response
Strike the sucface	Start a synthesized percussion sound.
Strike gentle or firmly	Change loudness using the relative vibration level.
Tilt the drumstick	Change pitch within a limited musical range.

The first sound will be a short electronic percussion tone. Once the input is reliable, I will test using the captured vibration to excite a resonator or Karplus–Strong style synthesis. This could retain more of the physical strike in the resulting sound. I will compare these approaches through recorded examples before choosing the final sound.

Initial scope

The core prototype will use a wired connection, USB power, one striking surface, and two interaction mappings. Daisy will generate the sound without a computer running the synthesis. Additional surfaces and a piezo inside the drumstick are possible extensions after the core instrument works reliably.

Hardware and signal flow

The piezo will be mounted behind the striking surface to capture vibration. Its signal will pass through a protective conditioning circuit into Daisy's audio input. The motion sensor will be fixed inside the drumstick grip and communicate with Daisy over I²C. Daisy will process the audio, read the motion data, and send its output to a powered speaker through a suitable audio connection.

Component	Planned quality and propose
Daisy seed	1 board for audio processing,synthesis, and sensor control.
Piezo contact microphone	1 element attached to the striking surface.
Piezo condicationing circuit	1 circuit with suitable op amp, protection, resistors, and capacitors.
Six axis IMU breakout	1 module, such as LSM6DSOX, inside the drumstick grip.
Protoboard and connectors	A soldered assembly with headers and secured cable connections.
Physical structure and audio	A striking panel, printed grip and enclosure, USB supply, audio connector, and powered speaker.

Circuit design and protoboard assembly

We will prepare a piezo and amplifier schematic and a motion sensor connection schematic. The piezo front end will provide a suitable input impedance, voltage protection, and level matching. Gain will be chosen from measured strike levels because a hard strike can overload the input. Biasing and coupling will be designed for the selected Daisy hardware revision rather than treated as a generic analog connection.And test the circuits before soldering them onto a protoboard. The drawings will identify component values, power and ground connections, connector pinouts, and Daisy pins. The final assembly will use sockets or headers where practical so that the boards can be removed for repair. TI's piezo application note and the official sensor and Daisy documentation will guide this work.

Drumstick and enclosure design

We will design and 3D print a drumstick grip or sleeve with a fixed sensor mount, a wire channel, and a removable cover. A separate enclosure will hold Daisy and the protoboard, with openings for USB and audio. Cable strain relief will reduce pulling at solder joints. I will test grip comfort, balance, and sensor movement before final printing. Keeping the main electronics in a separate box should make the drumstick easier to hold.

Development plan

Stage	Work in checkpoint
Input and Output	Generate a test tone; test piezo levels and motion readings separately.
Sound and mappings	Implement strike response and tilt control; compare synthesis approaches.

Physical build	Complete schematics, solder the protoboard, and fit the printed parts.
Test and Document	Check repeated playing, revise the design, and record the final demonstration.

Testing and expected outcomes

We will test soft and firm strikes at a consistent location, repeated strokes at different speeds, and the same tilt gesture several times. We will check for missed or duplicate sounds, clipping, unstable pitch changes, and loose connections. Motion readings will be smoothed to reduce changes caused by impact shocks. Vibration level will be treated as relative strike intensity, not a calibrated force measurement.

A successful prototype will respond consistently to individual strikes, provide an audible difference between gentle and firm playing, and let the user repeat a pitch change through tilting. We will record the physical strike and electronic response to assess delay. We will also invite a few classmates to try the instrument and describe which controls they understand without explanation.

The final submission will include the working instrument, firmware, both schematics, a protoboard wiring record, printable 3D files, and a short demonstration video. The documentation will explain the chosen sound mappings and changes made after testing.

Reference website analysis

The main reference is Preisler and Overholt's Hybrid Drum project [1]. It uses piezo input and Daisy Seed to connect physical percussion with digital synthesis. Its first Daisy version uses actual impact audio to excite Karplus–Strong synthesis; a later version adds battery operation and an internal actuator. The paper also identifies feedback instability and describes gyroscope software as unfinished. It provides a useful design reference, but not a complete soldering schematic. Our project will develop a drumstick based motion interface and a separate striking surface, using the reference to inform the audio path and iterative testing.